Assessing Ambiguity in Vision-Language Instructions

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Problem Statement

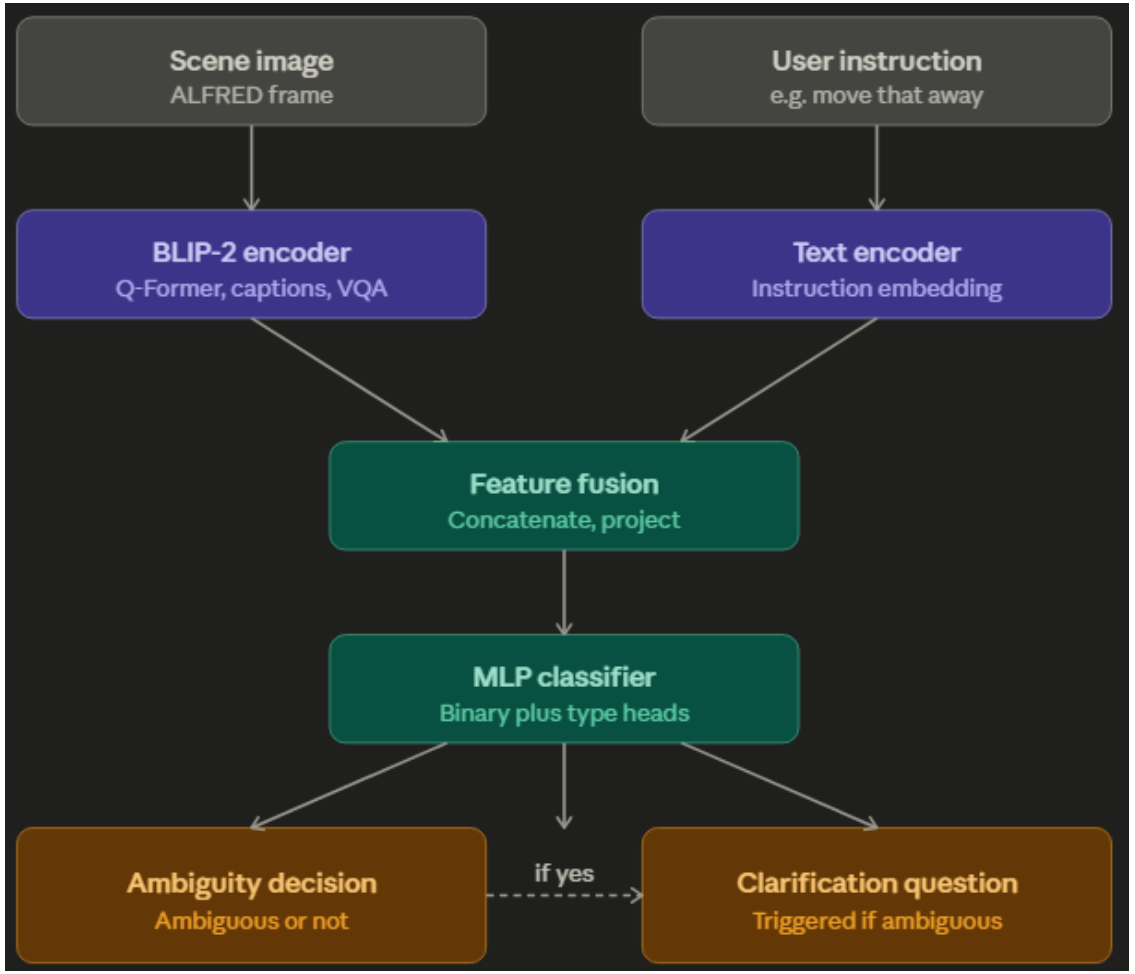
- Robots and embodied AI agents often receive natural language instructions that are underspecified
  - Example: "Move that away"
- Without knowing what "that" refers to, the instruction may be impossible to execute
- **Goal:** Detect when an instruction needs clarification, using both the user instruction and the scene image

Key Challenges

- Natural language ambiguity depends on visual context
  - The same instruction can be clear or ambiguous depending on the scene
- Text-only models miss visual cues that determine whether references like "that" or "there" are resolvable
- DialFRED provides ambiguity labels
  - Problem: linking them to ALFRED scene images requires non-trivial alignment
- Class imbalance: most DialFRED examples are labeled ambiguous, biasing the classifier

Approach/Solution

- BLIP2 encoder: strong visual grounding without full fine-tuning
- Caption + VQA probing
- Sentence-transformer text embedding
- MLP Fusion classifier



Novelty

- Combines DialFRED's ambiguity labels with ALFRED's visual frames
  - Most prior work uses one or the other
- Uses BLIP-2 caption variance and VQA probing as explicit ambiguity signals, not just embeddings
- Lightweight trainable head over frozen encoders
  - Avoids expensive full multimodal training
- Outputs both an ambiguity decision and a clarification question, rather than detection alone

Method details

Inputs

- Instruction text + scene image (ALFRED frame)

Feature extraction:

- BLIP-2 Q-Former image embedding
- Multiple generated captions (caption variance signal)
- VQA probing answers ("What object is referenced?", etc.)
- Sentence-transformer instruction embedding

Classifier

- Concatenated multimodal feature vector
- MLP with ReLU, batch normalization, dropout
- Binary output: ambiguous vs. not ambiguous

Dataset and evaluation metrics

DATASET

DialFRED + ALFRED

DialFRED — Instruction labels

QA annotation rows	Unique examples
53,004	4,574

- Labels from whether annotators marked clarification as necessary
- Aggregated from QA rows to instruction level

AMBIGUITY TYPES

location | appearance | direction | other

ALFRED — Visual scenes

Trajectory JSON files
8,051

- Embodied household tasks across train, val seen, val unseen, test
- Raw image frames from agent trajectories
- Provides visual scene observations for the model

ALIGNED BY: split | task ID | trial ID | subgoal timing

DialFRED label distribution — Instruction-level

Ambiguous	Not ambig.
~85.5%	~14.5%

Heavy class imbalance. The ambiguous class dominates by roughly 6:1, which shapes how the model trains and how we read per-class metrics downstream.

Metrics

- Accuracy
- F1 score
- Precision
- Recall
- Confusion matrix
- Qualitative clarification quality

Results

MULTIMODAL MLP

Larger aligned subset

Accuracy	Macro F1	Precision	Recall
0.74	0.72	0.73	0.74

Per-class metrics

	P	R	F1
Not ambiguous	0.76	0.66	0.71
Ambiguous	0.72	0.82	0.77

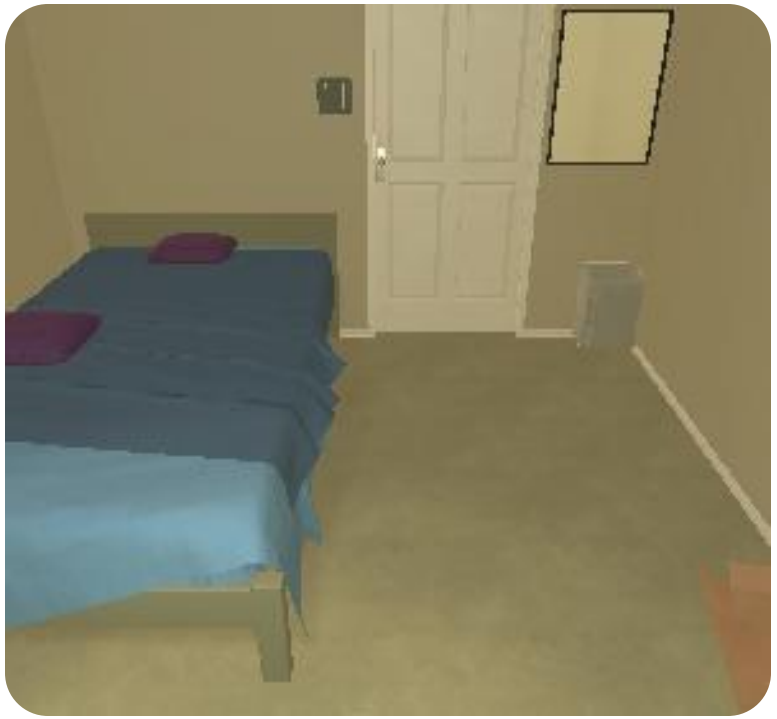
Confusion matrix

	Pred Not ambig.	Pred Ambig.
Not ambig.	66	34
Ambig.	18	82

Qualitative Results

QUALITATIVE EXAMPLE

Confidently correct prediction



INSTRUCTION split: train

"Go to the basketball, pick up the basketball"

BLIP-2 CAPTION cap. ambig. **0.656**

"a 3d rendering of a bedroom with a bed and a door"

TRUE
Not ambiguous

PRED
Not ambiguous

P(AMBIGUOUS) **0.131**

0 0.5 threshold 1

AMBIGUITY TYPE location

Discussion/Analysis

- The text-only model achieves strong overall F1, but performance is affected by class imbalance.
  - Struggles more with non-ambiguous cases
- The multimodal pipeline improved ambiguity detection because visual context can reveal whether references like "that", "there", or "it" are resolvable.
- Current implementation supports the full image + text pipeline.